

## LUBBOCK, TX, USA

WMO: 722670

Lat: 33.666N

Lon: 101.823W

Elev: 3254

StdP: 13.05

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23042

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 15.9       | 20.1 | -0.5                          | 6.0 | 30.7 | 3.6 | 7.5 | 37.3 | 33.6                  | 52.6 | 30.3 | 55.6 | 9.9                   | 30   | 0.616 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 23.1                   | 99.7            | 66.2 | 97.3 | 67.0 | 94.9 | 67.0 | 72.5                | 87.9 | 71.4 | 86.9 | 70.6 | 85.9 | 14.2                 | 170  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.4                            | 118.1 | 76.9 | 67.4 | 113.8 | 75.9 | 66.4 | 110.0 | 75.6 | 38.3          | 87.3 | 37.3 | 87.0 | 36.5 | 85.8 | 77.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |       |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |       |
|                   |      |      | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |       |
| 29.2              | 26.1 | 23.6 | DB                         | 9.3 | 104.8              | 4.3 | 3.9                                                | 6.2 | 107.6      | 3.7 | 109.9      | 1.4 | 112.2      | -1.7 | 115.0 |
|                   |      |      | WB                         | 7.7 | 74.7               | 4.1 | 1.4                                                | 4.8 | 75.7       | 2.4 | 76.5       | 0.1 | 77.2       | -2.9 | 78.2  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 61.6   | 41.4  | 45.2  | 53.0  | 61.2  | 70.4  | 78.8  | 81.1  | 79.9  | 72.6  | 61.9  | 50.6  | 41.6 |
|                                                                               | DBStd       | 16.49  | 9.13  | 10.16 | 9.87  | 8.87  | 8.36  | 5.84  | 4.53  | 4.66  | 6.46  | 8.50  | 9.01  | 9.11 |
|                                                                               | HDD50       | 980    | 289   | 191   | 83    | 17    | 1     | 0     | 0     | 0     | 0     | 14    | 101   | 284  |
|                                                                               | HDD65       | 3217   | 730   | 555   | 384   | 170   | 44    | 1     | 0     | 0     | 16    | 158   | 435   | 724  |
|                                                                               | CDD50       | 5197   | 23    | 56    | 176   | 353   | 633   | 863   | 965   | 926   | 677   | 382   | 118   | 25   |
|                                                                               | CDD65       | 1959   | 0     | 0     | 12    | 55    | 211   | 414   | 500   | 461   | 242   | 61    | 3     | 0    |
|                                                                               | CDH74       | 23384  | 12    | 72    | 315   | 941   | 2627  | 4869  | 5993  | 5214  | 2472  | 758   | 102   | 9    |
|                                                                               | CDH80       | 11645  | 0     | 11    | 73    | 330   | 1295  | 2636  | 3231  | 2726  | 1083  | 244   | 16    | 0    |
| Wind                                                                          | WSAvg       | 11.9   | 11.7  | 12.4  | 13.5  | 14.4  | 13.5  | 13.1  | 10.7  | 9.8   | 9.9   | 11.0  | 11.3  | 11.2 |
| Precipitation                                                                 | PrecAvg     | 18.7   | 0.6   | 0.7   | 0.9   | 1.1   | 2.5   | 2.7   | 2.2   | 2.1   | 2.5   | 1.8   | 0.8   | 0.7  |
|                                                                               | PrecMax     | 33.3   | 2.8   | 2.5   | 5.9   | 5.8   | 12.1  | 8.5   | 7.2   | 8.9   | 8.7   | 10.8  | 6.7   | 2.2  |
|                                                                               | PrecMin     | 5.9    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0  |
|                                                                               | PrecStd     | 5.5    | 0.6   | 0.6   | 1.1   | 1.1   | 2.0   | 1.8   | 1.8   | 1.7   | 2.2   | 2.1   | 1.1   | 0.6  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 75.7  | 81.3  | 86.4  | 91.3  | 100.3 | 104.9 | 102.3 | 100.9 | 97.0  | 90.7  | 82.2  | 74.9 |
|                                                                               |             | MCWB   | 49.8  | 53.0  | 55.4  | 56.9  | 61.8  | 64.6  | 67.3  | 68.0  | 66.7  | 61.5  | 55.6  | 50.4 |
|                                                                               | 2%          | DB     | 70.2  | 75.6  | 81.6  | 87.6  | 96.1  | 99.9  | 98.7  | 97.4  | 93.0  | 86.2  | 76.8  | 69.1 |
|                                                                               |             | MCWB   | 48.2  | 50.6  | 53.5  | 56.2  | 61.1  | 65.5  | 68.0  | 68.4  | 66.1  | 61.9  | 54.3  | 48.1 |
|                                                                               | 5%          | DB     | 64.4  | 70.5  | 77.4  | 84.0  | 92.0  | 96.5  | 96.5  | 95.1  | 90.1  | 82.3  | 72.5  | 63.7 |
|                                                                               |             | MCWB   | 45.6  | 49.1  | 52.5  | 56.0  | 61.1  | 66.5  | 68.2  | 68.4  | 66.2  | 60.8  | 52.9  | 46.4 |
|                                                                               | 10%         | DB     | 59.1  | 64.4  | 72.3  | 79.7  | 87.7  | 93.2  | 93.9  | 92.7  | 86.6  | 78.0  | 67.5  | 58.6 |
|                                                                               |             | MCWB   | 43.6  | 46.4  | 51.4  | 55.3  | 61.0  | 66.5  | 68.4  | 68.6  | 66.0  | 59.3  | 51.3  | 44.6 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 53.4  | 56.9  | 61.7  | 65.8  | 70.1  | 73.3  | 74.1  | 73.8  | 71.6  | 69.4  | 61.9  | 55.8 |
|                                                                               |             | MCDB   | 66.2  | 67.9  | 70.2  | 77.2  | 82.7  | 90.1  | 87.8  | 88.6  | 85.3  | 79.0  | 69.9  | 63.4 |
|                                                                               | 2%          | WB     | 50.0  | 53.4  | 59.1  | 63.2  | 68.2  | 71.7  | 72.4  | 72.4  | 70.3  | 67.0  | 58.4  | 51.1 |
|                                                                               |             | MCDB   | 65.3  | 68.1  | 69.6  | 73.4  | 81.5  | 87.3  | 87.7  | 87.8  | 83.5  | 76.6  | 68.1  | 63.8 |
|                                                                               | 5%          | WB     | 47.4  | 50.8  | 56.5  | 60.9  | 66.7  | 70.6  | 71.2  | 71.3  | 69.1  | 64.8  | 56.0  | 48.1 |
|                                                                               |             | MCDB   | 62.4  | 67.1  | 69.3  | 72.6  | 80.2  | 85.9  | 87.3  | 87.0  | 82.1  | 73.9  | 66.5  | 61.9 |
|                                                                               | 10%         | WB     | 44.4  | 48.0  | 53.5  | 58.6  | 65.3  | 69.5  | 70.3  | 70.3  | 68.1  | 62.7  | 53.4  | 45.1 |
|                                                                               |             | MCDB   | 58.4  | 63.8  | 68.9  | 71.5  | 78.5  | 84.4  | 86.2  | 85.8  | 80.9  | 72.1  | 65.0  | 58.0 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 27.2  | 27.9  | 28.4  | 28.2  | 26.3  | 24.8  | 23.1  | 23.2  | 23.6  | 25.7  | 26.4  | 25.8 |
|                                                                               |             | MCDBR  | 36.8  | 37.6  | 35.2  | 33.7  | 31.7  | 29.1  | 26.2  | 25.7  | 28.0  | 30.7  | 33.5  | 34.7 |
|                                                                               |             | MCWBR  | 19.6  | 18.9  | 15.6  | 13.9  | 12.1  | 8.9   | 7.8   | 7.8   | 9.2   | 12.5  | 16.1  | 18.7 |
|                                                                               | 5% WB       | MCDBR  | 33.0  | 33.7  | 29.2  | 27.2  | 25.9  | 24.7  | 22.9  | 22.7  | 22.3  | 22.7  | 26.8  | 30.6 |
|                                                                               |             | MCWBR  | 18.3  | 18.5  | 14.7  | 14.3  | 12.0  | 9.0   | 7.8   | 7.6   | 8.1   | 10.9  | 14.5  | 17.8 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.260  | 0.271 | 0.297 | 0.331 | 0.342 | 0.382 | 0.387 | 0.374 | 0.347 | 0.317 | 0.288 | 0.267 |      |
|                                                                               | taud        | 2.580  | 2.560 | 2.496 | 2.390 | 2.424 | 2.355 | 2.391 | 2.386 | 2.443 | 2.503 | 2.517 | 2.556 |      |
|                                                                               | Ebn at Noon | 309    | 315   | 312   | 303   | 298   | 285   | 283   | 286   | 291   | 293   | 294   | 299   |      |
|                                                                               | Edh at Noon | 24     | 28    | 32    | 38    | 37    | 40    | 38    | 37    | 34    | 29    | 26    | 24    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1059   | 1308  | 1705  | 2093  | 2247  | 2389  | 2290  | 2107  | 1744  | 1387  | 1105  | 918   |      |
|                                                                               | RadStd      | 101    | 123   | 124   | 143   | 177   | 138   | 112   | 94    | 119   | 157   | 113   | 72    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (14 neighbors) | +0.56 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +198  | +131  |

Nomenclature: See separate page